BloodBridge: Blood Donation & Emergency Request System

AI-Powered Blood Donation & Emergency Request Platform

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Executive Summary

Overview of BloodBridge

BloodBridge is an AI-powered web application designed to streamline blood donation and emergency request processes. It connects donors, recipients, and hospitals through a centralized digital platform, ensuring faster response times and improved accessibility during critical situations.

Problem Statement

Traditional blood donation systems suffer from inefficiencies such as delayed donor matching, lack of centralized data, and poor communication during emergencies. These challenges often result in life-threatening delays.

Proposed Solution

BloodBridge introduces a real-time, AI-driven platform that connects donors and recipients instantly. It leverages geolocation, intelligent donor matching, and automated notifications to ensure timely blood availability.

Key Benefits

- Instant donor-recipient connection
- Centralized donor database
- Real-time emergency alerts
- Enhanced donor engagement
- Secure and scalable architecture

Introduction

Background

Blood donation is a critical healthcare service that saves millions of lives annually. However, traditional systems rely heavily on manual coordination, leading to inefficiencies and delays.

Importance of Blood Donation

Regular blood donation ensures a stable supply for hospitals and emergency cases. A digital platform can significantly improve donor participation and accessibility.

Challenges in Traditional Systems

- Manual donor search and verification
- Lack of real-time communication
- Limited donor engagement
- Fragmented data management

Vision of BloodBridge

To create a unified, intelligent, and responsive platform that revolutionizes blood donation and emergency response through technology.

Problem Statement

- Delays in finding compatible donors
- Lack of centralized donor information
- Poor communication during emergencies
- Low donor engagement and retention

Objectives

- Connect donors and recipients instantly
- Enable emergency blood requests
- Search donors by blood group and location
- Increase donor participation
- Improve response time and coordination

Proposed Solution

Platform Overview

BloodBridge is a web-based system that integrates donor registration, recipient requests, and admin management into a single platform.

User Workflow

1. Register and log in
2. Update profile and availability
3. Search for donors or create a blood request
4. Receive notifications and respond

Donor Workflow

- Register and verify blood group
- Set availability status
- Receive and respond to requests

Recipient Workflow

- Register and submit blood requests
- Search for nearby donors
- Track request status

Admin Workflow

- Manage users and requests
- Verify donor authenticity
- Monitor system analytics

Features

- **User Authentication:** Secure login and registration using JWT.
- **Donor Registration:** Capture donor details, blood group, and location.
- **Recipient Registration:** Enable patients to request blood.
- **Blood Request Management:** Track and manage requests in real time.
- **Donor Search:** Filter donors by blood group and proximity.
- **Emergency Alerts:** Notify nearby donors instantly.
- **User Dashboard:** Personalized dashboard for donors and recipients.

- **Profile Management:** Update personal and medical details.
- **Availability Status:** Toggle donor availability.
- **Admin Panel:** Manage users, requests, and reports.

Technology Stack

Category	Technologies
Frontend	React, Vite, Tailwind CSS, Axios
Backend	Spring Boot, Spring Security, JWT Authentication, Spring Data JPA, Hibernate
Database	MySQL
Development Tools	Maven, Git, GitHub, Postman, VS Code, IntelliJ IDEA

System Architecture

Architecture Diagram:
User → React Frontend → REST API → Spring Boot Backend → MySQL Database

Description:

- The frontend communicates with the backend via REST APIs.
- The backend handles authentication, business logic, and database operations.
- MySQL stores user, donor, and request data securely.

Database Design

Entities:

- **User:** Stores login credentials and roles.
- **Donor:** Contains donor-specific details.
- **Blood Request:** Tracks requests and statuses.
- **Admin:** Manages system operations.

ER Diagram:
User (1) → Donor (1)
User (1) → Blood Request (M)
Admin (1) → Blood Request (M)

Module Description

Authentication Module

Handles user registration, login, and JWT-based authentication.

Donor Module

Manages donor profiles, availability, and request responses.

Recipient Module

Allows recipients to create and track blood requests.

Search Module

Implements donor search by blood group and location.

Blood Request Module

Processes and monitors blood requests and notifications.

Admin Module

Provides administrative control over users and system data.

API Overview

Endpoint	Method	Description	Auth Required
/api/auth/register	POST	Register new user	No
/api/auth/login	POST	Authenticate user	No
/api/donors	GET	Fetch donor list	Yes
/api/donors/{id}	GET	Get donor details	Yes
/api/requests	POST	Create blood request	Yes
/api/requests/{id}	GET	View request details	Yes
/api/admin/users	GET	Manage users	Admin
/api/admin/requests	GET	View all requests	Admin

User Interface

Screens (Placeholder Previews):

- Home Page
- Login Page
- Registration Page
- Dashboard

- Donor Search
- Blood Request Form
- User Profile

Each screen follows a red, white, and dark gray theme with clean typography and healthcare icons.

Workflow Diagram

Flow:
Registration → Login → Search Donor → Send Request → Donor Notification → Accept Request → Successful Blood Donation

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Security Features

- **JWT Authentication:** Secure token-based access.
- **Password Encryption:** Encrypted credentials using BCrypt.
- **Role-Based Access:** Separate permissions for users and admins.
- **Secure API Communication:** HTTPS and token validation.
- **Input Validation:** Prevents SQL injection and XSS attacks.

Future Enhancements

- AI-based donor recommendation system
- Real-time location tracking
- Hospital and blood bank integration
- SMS and email alerts
- Mobile application
- AI chatbot for assistance
- Predictive analytics for blood demand

Project Timeline

Phase	Duration	Description
Planning	Week 1–2	Requirement gathering and analysis
Design	Week 3–4	UI/UX and architecture design
Development	Week 5–10	Frontend and backend implementation
Testing	Week 11–12	Unit, integration, and user testing
Deployment	Week 13	Final deployment and documentation

Conclusion

BloodBridge transforms the traditional blood donation process into a fast, intelligent, and reliable digital system. By integrating AI, real-time communication, and secure data management, it ensures that every drop of blood reaches those in need without delay.

References

- Spring Boot Documentation – spring.io/projects/spring-boot
- React Documentation – react.dev
- MySQL Documentation – mysql.com
- JWT Authentication – jwt.io
- REST API Design – restfulapi.net